

Review retrospective:
*the anharmonic potential and its coercivity — and a
foundation-file import cleared by a full build*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/OneD/Anharmonic.lean`: the base potential $L(x) = (\lambda/2)x^2 + (\alpha/6)x^3 + (\gamma/24)x^4$, its decomposition into harmonic plus perturbation, and the coercivity result $cx^2 \leq L(x)$ under the discriminant condition $\alpha^2 < 3\lambda\gamma$ — the fact the entire 1D anharmonic arc consumes. Result: **a clean fidelity pass with three dead-code removals**, the most interesting of which (a genuinely-unused import on a foundation file) was cleared by a full project build rather than a single-module rebuild.

1 Setting

This is the base of the 1D anharmonic programme (primer Stage 2 / `lem:laplace_cov2`). Everything downstream — the partition derivatives, the cumulant ladder, the FDT asymptotics, and the rescaling file reviewed just before — ultimately rests on `anharmonic_coercive`. Reviewing it after its consumers closes the 1D anharmonic foundation from the top down.

2 What was reviewed

Two definitions (the potential and the cubic+quartic perturbation) and seven lemmas/theorems: the harmonic+perturbation decomposition (pointwise and functional), the perturbation’s vanishing and x^3 factoring, the Boltzmann-factor split, the factored form $L = x^2Q$, the quadratic minimum, and the two coercivity statements (quadratic lower bound, then $cx^2 \leq L$).

3 Fidelity / soundness findings

Sound; all the algebra checks out. The coefficients match the primer; the factored form $L = x^2((\gamma/24)x^2 + (\alpha/6)x + \lambda/2)$ is the correct regrouping; and the load-bearing square-completion is right: $Q(x) - \frac{3\lambda\gamma - \alpha^2}{6\gamma} = \frac{\gamma}{24}(x + 2\alpha/\gamma)^2$, using $\frac{3\lambda\gamma - \alpha^2}{6\gamma} = \frac{\lambda}{2} - \frac{\alpha^2}{6\gamma}$, so the coercivity constant $c = \frac{3\lambda\gamma - \alpha^2}{6\gamma}$ is positive exactly under $\alpha^2 < 3\lambda\gamma$.

Two benign notes, recorded but not escalated and not acted on, both of the “the statement is sound and is the form the consumers want, the docstring prose just says a little more” kind:

- `anharmonic_quadratic_min` proves the lower-bound half ($\frac{3\lambda\gamma - \alpha^2}{6\gamma} \leq Q(x)$), which is all coercivity needs; the docstring’s “attained at $x = -2\alpha/\gamma$ ” is accurate but not separately formalised.

- `anharmonic_coercive` is the existential $\exists c > 0$, which is exactly what the downstream consumes (the rescaling file does `obtain ⟨c, ...⟩`); the explicit witness named in the docstring is recoverable from the proof but not part of the statement.

4 Simplifications applied

Three dead-code removals. A dead `open Real` (every `exp` in code is `Real.exp`; the only bare `exp` is in comments), a semantically-inert `rw [show (c*x^2) = c*x^2 from rfl]` no-op in the coercivity proof, and — the interesting one — a genuinely-unused `import Laplace.OneD.Localisation` (this file references nothing from `Localisation`, which exports only `harmonic_tail_*` lemmas).

The import is the point worth dwelling on. This is a *foundation* file: the whole `anharmonic` arc imports it, and `Localisation` is *not* re-provided by the other import (`Harmonic`), so removing it shrinks `Anharmonic`’s import closure. A single-module `lake build Laplace.OneD.Anharmonic` would say nothing about whether some consumer was relying on obtaining `Localisation` *transitively through* `Anharmonic`. So the removal was arbitrated by a **full project build**: `lake build` went green at 8306 jobs (unchanged from baseline), with every consumer — `AnharmonicGibbsRegularity`, `HarmonicCrossSuscDeriv`, and the rest of the arc — rebuilding successfully. That is the definitive evidence that no downstream file depended on the transitive import. All nine declarations stayed byte-identical; GPT-affirmed; fast-forward-merged.

GPT’s remaining suggestions were declined: the `_hlam` parameter is already underscore-marked as intentionally unused (and is signature-frozen anyway), and the rest were tactic-golf (a `simp` for the commutativity rewrite, compressing the `exp_add` step, tightening the square-completion proof).

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **For a redundant import on a foundation file, the full build is the honest test — and it is worth running.** The single-module rebuild that suffices for a dead `open` (file-local) or a leaf-file import is *not* enough when the file has consumers and the import is not re-provided by a sibling: a consumer could compile only because the import reached it transitively. A green full build settles it conclusively, and clears the hazard that kept import-removal out of scope on the Multi/Basic review.
- **Reviewing a foundation after its consumers is the right order for an arc.** By the time this file was reviewed, its dependents were already audited; confirming the base is sound (and trimming its cruft under a full build) closes the arc with confidence flowing both directions.

7 Follow-ups

- The 1D non-cumulant layer still has tractable never-reviewed files (`Harmonic.lean`, `Localisation.lean`, the J-function and `Quartic/Sextic/Monomial` families); the `Multi/Covariance*` trio remains the large sectioned-approach target.